

IF2RNA

Predicting Spatial Gene Expression from GeoMx Immunofluorescence Images

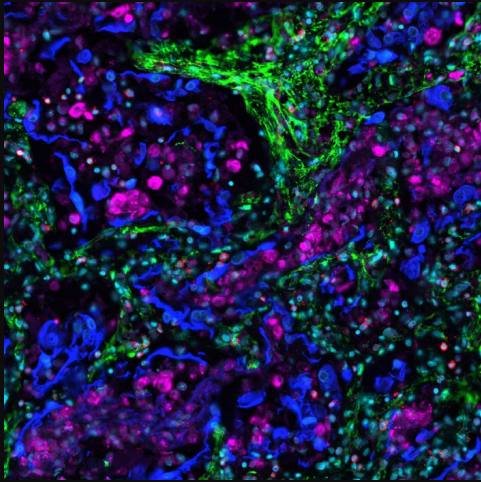
By Siddarth Chilukuri

Problem and Motivation

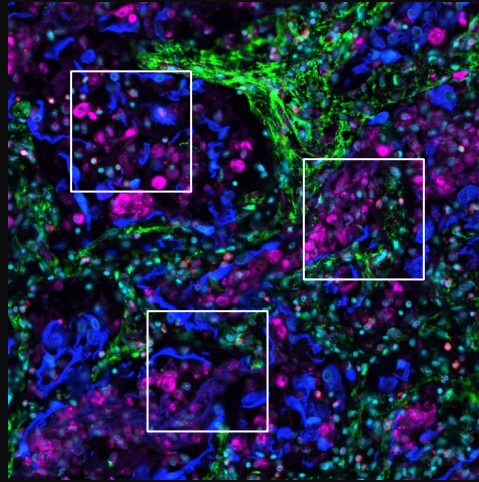
Problem and Motivation

- GeoMx gives us RNA measurements for selected ROIs, but not dense expression across the full slide
- ROI selection is sparse and can miss smaller or less obvious microenvironments
- IF morphology is already available in the workflow, so it is a natural signal to try to use
- The goal is to predict RNA from IF and eventually support dense virtual expression maps

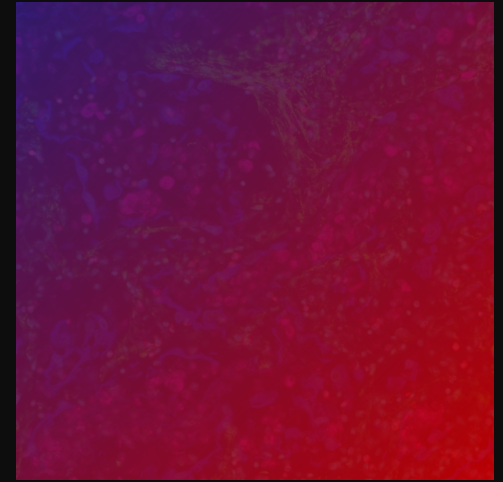
Problem and Motivation



IF Slide



Selected ROIs



Dense Virtual
Map

Related Work

- Prior work predicts RNA from images (e.g., HE2RNA)
- Often uses slide-level or grid-based supervision
- Our setting is harder: ROI-level, irregular shapes
- So we use attention MIL for weak supervision

Setup and Methods

Data and Experimental Setup

- Input: ROI-level IF-derived features
- Target: matched RNA expression vectors for the same ROIs
- Main benchmark: in-distribution evaluation with patient-level splits
- Stress tests: held-out organ and held-out slide evaluation

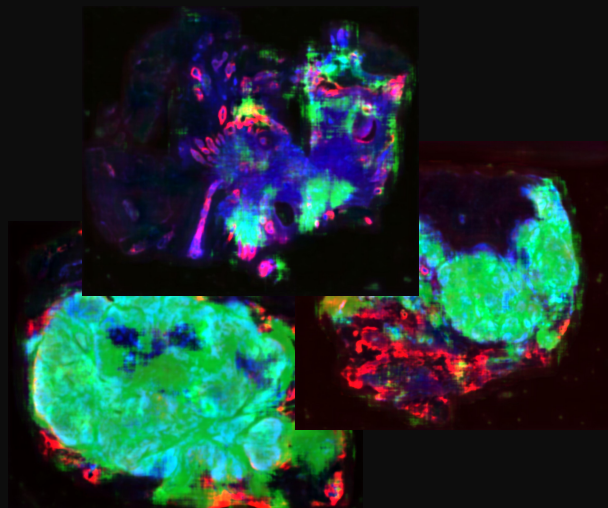
Data and Experimental Setup

- Compared models: Attention MIL, ViS, ViT, and a MeanPool Ridge baseline
- Metrics: mean and median gene-wise Pearson r , MAE, RMSE, and counts above useful correlation thresholds

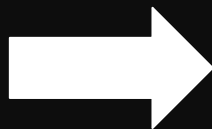
Method Overview

- MeanPool Ridge: simple baseline on pooled ROI features
- ViS / ViT: neural set or sequence encoders over ROI feature bags
- Attention MIL: learns which instances within an ROI should matter more
- Same training and evaluation framework used across all methods

How the Attention MIL model works



ROI Patch Bag



ResNet-50 Patch Encoder

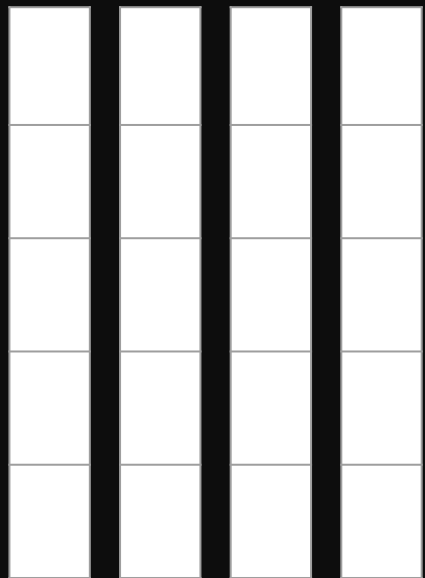
How the Attention MIL model works

ResNet-50 Patch Encoder



ROI Patch Bag

How the Attention MIL model works



ROI Patch Bag



Attention Pooling

How the Attention MIL model works

Attention Pooling



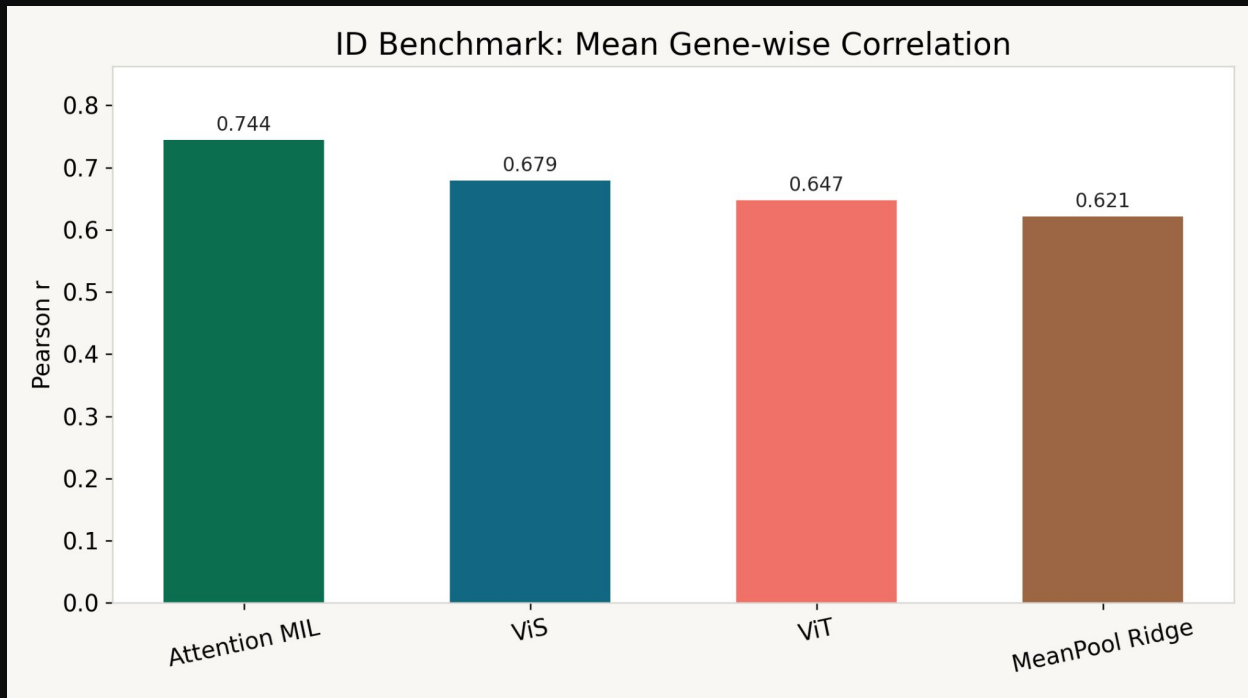
RNA Prediction

Results

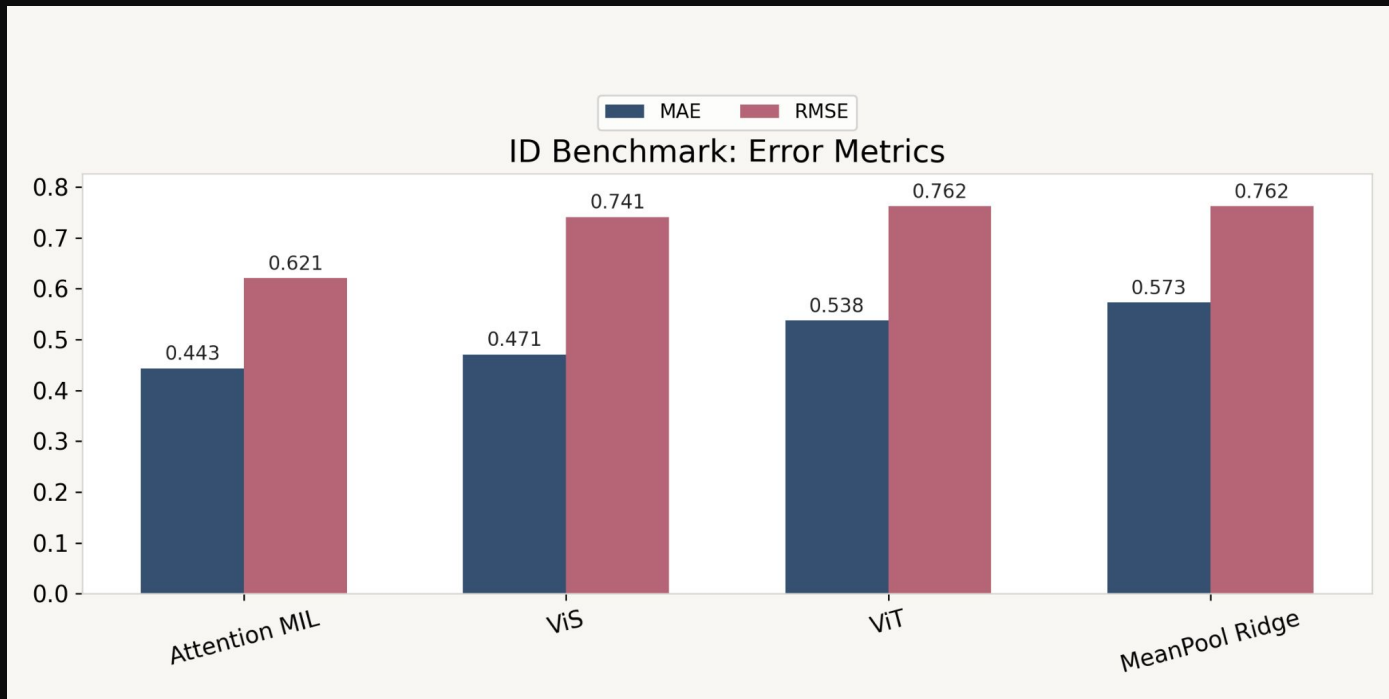
Main in-distribution benchmark

- Attention MIL is currently the strongest overall model on the ID benchmark
- It leads on both mean and median gene-wise correlation
- It also improves error metrics relative to the neural baselines in this snapshot
- MeanPool Ridge remains a useful anchor, but the best neural model is clearly ahead

Main in-distribution benchmark



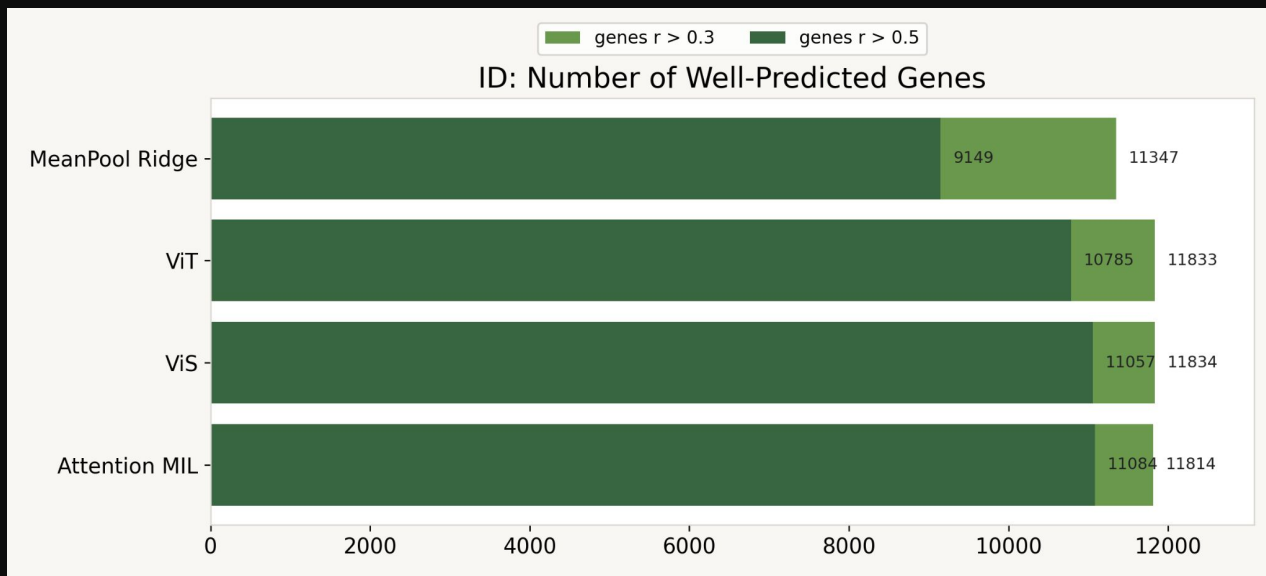
Main in-distribution benchmark



Performance across genes

- Average correlation is useful, but it can hide whether performance is concentrated in a small subset of genes
- So I also looked at how many genes pass practical correlation thresholds
- Attention MIL predicts a large number of genes above $r > 0.3$ and $r > 0.5$
- This suggests the gains are broad, not coming from only a few easy targets

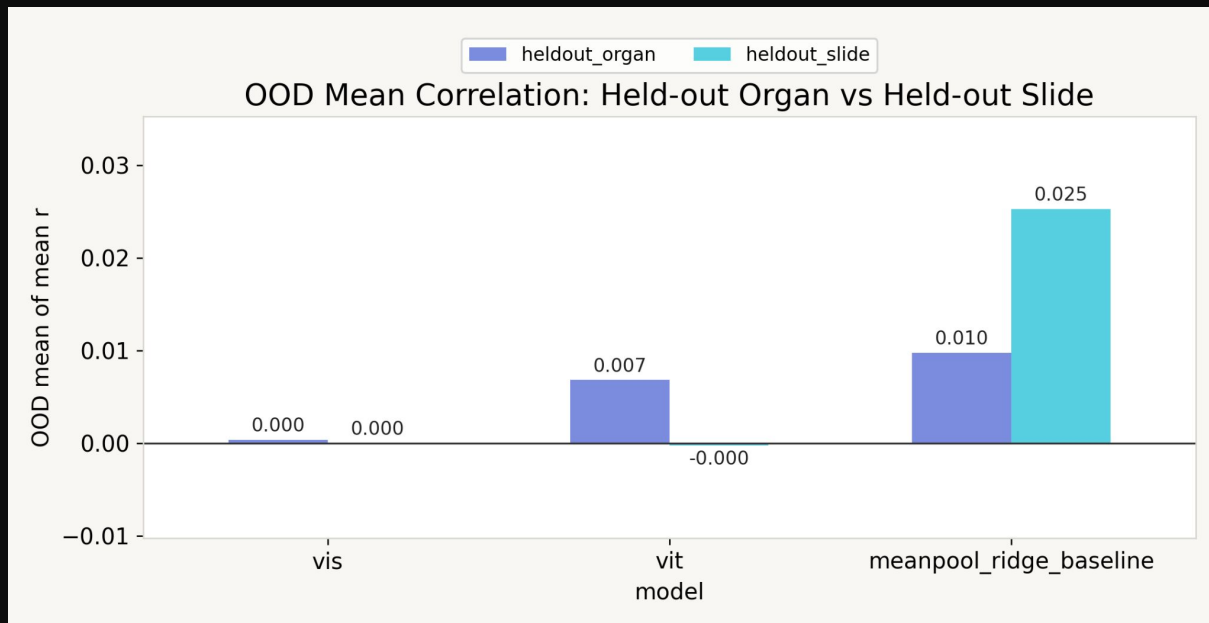
Performance across genes



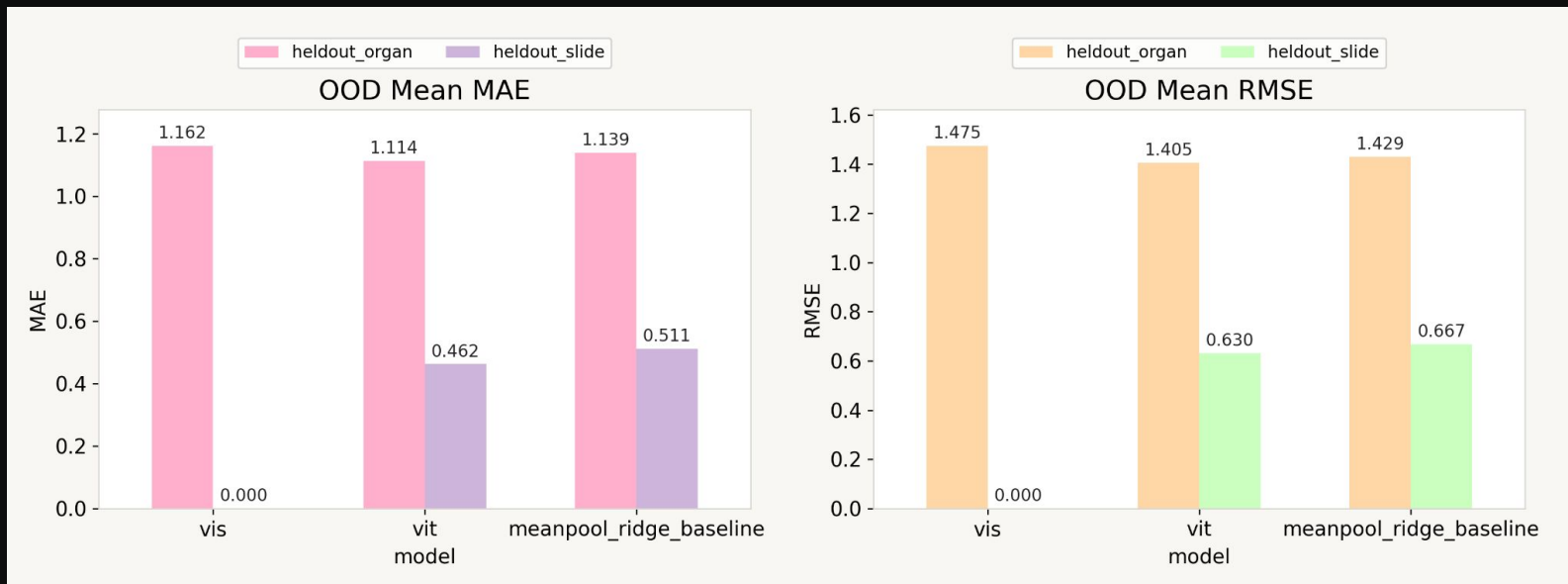
Out of Distribution Results

- Strong ID performance does not translate to OOD generalization
- I evaluated two OOD settings: held-out organ and held-out slide
- Both are much harder than the ID benchmark
- Correlations drop sharply and are often close to zero
- So the main weakness is not fitting the training distribution, but generalizing beyond it

Out of Distribution Results



Out of Distribution Results



Current Status and Next Steps

Key takeaways and next steps

- Predicting RNA from IF works well in-distribution
- Attention MIL is the strongest model in the current setup
- Generalization is the main limitation right now
- The model clearly learns something meaningful, but robustness is still the main challenge.

Key takeaways and next steps

- Next Steps:
 - Push OOD evaluation further (more complete organ and slide splits)
 - Look at organ-specific behavior more explicitly
 - Try simpler ways to improve robustness (e.g., normalization, feature constraints, maybe ensembling)

Thank you!